Resistance to Carcinogenesis in the Spiny Mouse (Acomys) correlates with upregulation of multiple tumor suppressor genes

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Article

Keywords:

Posted Date: June 4th, 2025

DOI: https://doi.org/10.21203/rs.3.rs-6221361/v1

Additional Declarations: No competing interests reported.

Version of Record: A version of this preprint was published at Scientific Reports on May 2nd, 2026. See the published version at https://doi.org/10.1038/s41598-026-45001-6.

Abstract

Cancer remains a leading cause of morbidity and mortality worldwide, and the relationship between cancer and regeneration remains poorly understood. The Spiny Mouse (*Acomys sp.)* has attracted considerable attention due to its regenerative abilities. In this study, we compared the response of *Mus musculus (C57BL6)* and *Acomys dimidiatus* mice to the DMBA/TPA papilloma inducing protocol. While both *Mus* and *Acomys* mice experienced carcinogenic damage to their skin cells, mounted proliferative responses and underwent immune cell infiltration, only *Mus*mice developed tumors, whereas *Acomys* remained tumor-free. To explore the molecular mechanisms underlying this resistance, we performed RNA sequencing on tissue samples from both species at baseline and at multiple time points during the carcinogenesis protocol. The data reveal distinctly different transcriptional responses between both species. *Acomys* showed significant upregulation of immune related genes, including a set of tumor suppressor genes, while *Mus* seemed to focus its response on modifying epidermis structure and regulation of the cell cycle.

Introduction

Upon injury, organisms have two fundamental choices: regeneration or scarring. Regeneration involves recreation of the original tissue architecture, while scarring typically results in development of a fibrotic tissue [3]. Regeneration involves a mandatory proliferative response designed to provide the cellular mass necessary for tissue reconstruction, which, if not strictly regulated, could lead to uncontrolled proliferation and cancer. In regeneration, proliferation is orderly, and importantly, responds to termination signals; if these signals are overridden, cancer may ensue. Wound healing and regeneration share major molecular mechanisms and pathways overlap with cancer[4–9], to the point where it has been proposed that cancer is a wound that never heals[10,11]. Throughout the animal kingdom, species in every taxon have been reported to have low rates of tumorigenesis. Species mentioned in the literature include bats, naked mole rats, elephants, and whales [1]. Possible mechanisms hypothesized to be involved include lower somatic mutation rate, shorter telomeres, redundancy of tumor suppressors, a more efficient immune system, higher apoptosis rate and increased contact inhibition, among others[2]. One prevailing hypothesis suggests that high regenerative capacity tends to be associated with a low susceptibility to cancer, presumably due to trade-offs between both outcomes[12]. Planarians are capable of regeneration of entire organs even from small body fragments[13], a process driven by pluripotent stem cells known as neoblasts[14]; however, the extent of this regenerative capacity varies across planarian species. Planarians from the genus *Schmidtea* are capable of full body regeneration, while those in the genus *Dugesia* exhibit more restricted regenerative abilities. Interestingly, *Dugesia* species also show a tendency to develop spontaneous outgrowths[15] Clique ou toque aqui para introduzir texto.as well as after exposure to cadmium[16,17], while *Schimidtea* does not, suggesting a link between reduced regenerative capacity and increased cancer susceptibility[18]. Knock-down of the planarian ortholog of p53 results cell cycle dysregulation and impaired regeneration[19,20]. Urodeles, including axolotls, salamanders, and newts, are capable of regenerating multiple tissues and organ systems. Despite axolotls having been observed to spontaneously develop skin tumors[21], urodeles are thought to have low rates of tumorigenesis, and are resistant to cancer after carcinogen exposure[22]. Zebrafish can regenerate a range of tissues after injury[23]. While spontaneous tumor formation in zebrafish is rare[24,25] exposure to carcinogens induces vigorous tumor formation in these animals[26,27]. *Mus musculus* laboratory strains show varying degrees of cancer susceptibility; treatment with 7,12-dimethylbenz[a]anthracene (DBMA) followed by the proliferation inducer 12-O-tetradecanoylphorbol-13-acetate (TPA), induces skin carcinogenesis[28,29] indicates tumor promotion varies across strains (SENCAR > DBA/2 ≥ CD-1 > C3H/He >> C57BL/6)[30].

The most striking feature of *Acomys* is its exceptional regenerative capacity[31–37] . *Acomys* responds to wounding by mounting a regenerative response, in contrast with mammals generally, which repair their injuries by fibrotic scarring[37]. In response to full thickness 4 mm diameter ear pinna punch wounds (comprising 35% of the ear pinna surface), *Acomys* mounts a strong proliferative response causing the tissue to be re-established and organized into a histological structure that faithfully resembles the original tissue architecture. In contrast, *Mus* simply heals the border of the wound by fibrotic scarring[31,37]. Dorsal skin wounds in *Acomys* also repair through a regenerative response, re-establishing dermis, epidermis, epidermal appendages, innervation and vascularization, as well as adipose and muscle layers[38]. Acute wound models of *Acomys* of heart and kidney, show resistance to ischemic injury[39,40]. Injection of myotoxins into striated muscle, while initially causing severe cellular damage, are undetectable after 2 months[38]. None of these responses are observed in *Mus*. Furthermore, a complete spinal cord transection in *Mus* leads to permanent loss of bladder control and hindlimb sensory/motor function. In contrast, *Acomys* with fully transected spinal cords regain bladder control within three weeks and recover up to 60% of motor function within 60 days post-injury[35].

In our regeneration sutdies, we have consistently observed that *Acomys* initiates a robust proliferative response early in the healing process but never recorded uncontrolled proliferation or the development of spontaneous tumors in aged animals, despite their relatively long lifespan of six years. Neither are there any reports of cancer resistance in Acomys in the literature. These observations suggest that regenerative capacity is linked to specific mechanisms that regulate proliferation, potentially leading to cancer resistance as a secondary consequence. This observation prompted us to conduct a systematic evaluation of *Acomys*' cancer susceptibility. In this study, we investigated the susceptibility of the spiny mouse (*Acomys dimidiatus*), to chemically induced tumor formation using the DBMATPA model, a well-established experimental system used to study chemically induced skin carcinogenesis[28,29] particularly in *Mus musculus.* Our work shows that *Acomys dimidiatus* is resistant to a DMBA/TPA papilloma inducing protocol that induces papilloma's in C57BL6 and provides insights into the underlying protective mechanisms operating in *Acomys.*

Results

Cancer resistance in Acomys

To explore whether *Acomys*' regenerative capacity involves specific mechanisms of cancer resistance, we employed the two-stage DMBA/TPA epidermal cancer induction protocol[29,41]. Animals were treated with an initial dose of DMBA, a mutagen known to cause DNA double strand breaks, followed by regular injections of TPA to induce proliferation, as described in Figure 1A and Materials and Methods. Papillomas started appearing in *Mus* by week 13. After 30 weeks of the protocol, 5 out of 6 *Mus* (C57/Bl6) had developed multiple papillomas in the treated area (Figure 1B and D). The number of tumors in the *Mus* group after 30 weeks was 1, 1, 1, 3, 0 and 10, giving an average of 2,66 tumors/animal. In contrast, 6 out of 6 *Acomys* exhibited no sign of hyperplasia or tumor formation during the duration of the experiment (Mann-Whitney test, p = 0.009; Figure 1C and D). Figure 1E illustrates the average number of papillomas per animal over the course of the DMBA/TPA treatment.

First, we asked whether the initial DMBA treatment was inducing double strand breaks (DSBs) to similar levels in both species. Immunohistochemistry (IHC) for the p2AHx a marker of DNA double strand breaks (DSB) 24 h after DMBA application in *Mus* revealed an increase of 1.64 % positive nuclei in control animals to 16.1 % in treated animals (p = 0.04). This is a 9.8-fold increase indicating the DMBA treatment had indeed caused genotoxic damage. Notably, *Acomys* showed very similar p2Hax positive nuclei (16.8 % vs. 2.8 % in treated and control respectively (p = 0.08) for a 5.9-fold increase (Figure 2A and B), indicating that *Acomys* had received a similar level of genotoxic damage as *Mus*. Concurrently with presence of DSBs, we observed an immediate proliferative response in the epidermis of both species, as indicated by IHC against Ki67, a canonical proliferation marker. 24 h after DMBA treatment, Ki67 positive nuclei increased from 8.3 % in untreated *Mus* to 26.8 % in treated *Mus* (p = 0.005), while in *Acomys* we observed Ki67 positive nuclei increase from 11.3 % to 19.65 % (p = 0.04), i.e., a 3.22-fold increase in *Mus* compared to a 1.73-fold increase in *Acomys*. Therefore, both species show an immediate proliferative response to DMBA at 24hs (Figure 2C).

Having established that a relatively similar genotoxic insult resulted in clear difference in amount of papillomas observed in both species after completion of the protocol, we then explored whether complete absence of tumors in the Spiny Mouse could be attributed to early events occurring within the first month after treatment that effectively halt tumorigenesis in *Acomys*. To investigate this, we conducted a transcriptomic analysis at three different time points: 1 day (D1), 14 days (D14) and 28 days (D28) after the start of the protocol.

Transcriptional Profile one day after* DMBA *treatment

Samples from *Mus* and *Acomys* collected at D1 (24 hours after treatment with DMBA) were compared to samples from untreated mice harvested at D0 (N=4, Figure 3A). PCA analysis revealed expected clustering of the samples including the *Acomys* samples, which exhibited some heterogeneity. We hypothesize that this may be due to the partially outbred nature of the animals of our colony, as we avoid consanguineous matings in our colony. The PCA1 and PCA2 components explained 41.56% and 18.72% of variation in *Mus* and 41.19% and 22.41% in *Acomys,* respectively (Supplementary Figure 1). We queried differentially expressed genes (DEGs) (log2 =1) and p-adj ≤ 0.05. We comparatively examined genes exclusively upregulated in *Mus* but not in *Acomys* (414) and genes exclusively upregulated in *Acomys* but not in *Mus* (17) (Figure 3B, 3C and 3E). Only 11 genes were upregulated in both species (Figure 3C). To understand what biological processes (BP) were associated with the set of genes upregulated exclusively in each species, we conducted a Panther analysis. Upregulated *Mus* genes resulted in 138 enriched BP ontological categories, the genes of which could be grouped in the following biological themes: cell cycle and proliferation related processes (31.4%); processes related to epidermis structure and morphogenesis (7.1%); processes related to apoptosis (3.2%) and other processes (57.6%). In contrast, no BP ontological categories were significantly enriched for genes exclusively upregulated in *Acomys*, and only 3 BP ontological categories were enriched due to genes upregulated in both species (Figure 3D and 3H).

We then examined genes exclusively downregulated in *Mus* but not *Acomys* (304) or exclusively downregulated in *Acomys* but not *Mus* (5) (Figure 3B, 3E and 3F). No downregulated genes were shared between species (Figure 3F). To understand what BPs were associated with the set of genes upregulated exclusively in each species, we performed a Panther analysis. Downregulated *Mus* genes revealed 30 enriched BP ontological categories (Figure 3G), the genes of which could be organized in the following biological themes: cell cycle and proliferation related processes (6.1 %); processes related to epidermis structure and morphogenesis (10.7%); processes related to immunity (3.2%) and other processes (80.1%). In contrast, no BP ontological categories were significantly enriched for genes exclusively downregulated in *Acomys*, or genes downregulated in both species (Figure 3H).

A BP and KEGG pathway analysis conducted with Novomagic software (Novogene Europe) showed that response to the DMBA insult at 24 hours did not result in either BP categories or KEGG pathway enrichment involving downregulated genes in either *Mus* or *Acomys*. However, the analysis did identify several BP categories and KEGG pathways significantly enriched for upregulated genes in both species (Supplementary Figure 2). Therefore, the initial response to DMBA seems to be mediated by upregulation of specific gene sets in each species. We were particularly interested in the 17 genes upregulated in *Acomys* but not in *Mus*. Among this set we found several genes with intriguing functions related to tumorigenesis. *NQ01*, a NADP dehydrogenase (log2 upregulated 3.45-fold) is involved in detoxification and preventing formation of reactive oxygen species, possibly preventing cellular damage[42]. Two other genes with tumor suppressor functions exclusively upregulated in *Acomys* are *GNK1* (log2 8.04-fold) and *SPINK7* (log2 4.13-fold). Intriguingly, we found *CYP1A1*, a cytochrome P450 enzyme which is essential for the biotransformation of polycyclic aromatic hydrocarbons such as DMBA[43] upregulated log2 5.1-fold in *Acomys*, but only log2 3.1-fold in *Mus*) (Table 1).

Transcriptional Profile at Day 14

Mus and *Acomys* samples at D14 (14 days after treatment with DMBA followed by TPA treatment at D7, D9 and D11) were compared to untreated samples harvested at D0 (N=4). PCA analysis with the expected clustering of samples again showed some heterogeneity for *Acomys* samples, but overall, the

PCA1 and PCA2 components explained 53.02% and 19.73% of variation in *Mus* and 41.91% and 23.97% in *Acomys,* respectively (Supplementary Figure 1). We queried DEGs (log2 =2) and p-adj ≤ 0.05 and comparatively examined genes exclusively upregulated in *Mus* but not in *Acomys* (399) and genes exclusively upregulated in *Acomys* but not in *Mus* (303) (Figure 4A, 4B and 4D). Only 39 genes were upregulated in both species (Figure 4B). To understand what BPs were associated with the set of genes upregulated exclusively in each species, we performed a Panther analysis. Genes upregulated exclusively in *Mus* revealed 170 enriched BP ontological categories, while we found 464 BP ontological categories significantly enriched for genes exclusively upregulated in *Acomys*, and only 3 BP ontological categories enriched based on the 33 genes upregulated in both species (Figure 4C). Analysis of the distribution of genes related to the enriched BP categories revealed striking differences in how each species responded to the treatment at this timepoint. In *Mus*, the number of genes in enriched BP categories could be grouped in the following biological themes: cell cycle and proliferation related processes (37.8%); processes related to epidermis structure and morphogenesis (10.7%); processes related to apoptosis (1.1%) and other processes (50.4%). In contrast, BP categories enriched in *Acomys* were related to immune response (29.9%), apoptosis (2%), cell cycle (0.6%), processes related to epidermic structure and morphogenesis (3.5%) and other processes (63.9%; Figure 4C and 4G). We examined the identity and known functions of genes upregulated exclusively in *Acomys* vs exclusively in *Mus*. Interestingly, we found that *Acomys* upregulated a total of 29 genes related to tumor suppression, with an average log2 fold increase of 4.31. Among these, several were upregulated to levels higher than log2 fold change = 6: *AHRR* (6.46); *CD89* (7.4); *CLCA2* (6.98); *GKN1* (6.78): *IL25* (8.41); *SHISA* (6.37); *OAS3* (8.84) and *ST18* (6.37). In contrast, we found that *Mus* upregulated only 10 genes related to tumor suppression, and to an average log2 fold level of 3.04 (Table 2).

Similarly, we examined genes exclusively downregulated in *Mus* but not in *Acomys* (80) and genes exclusively downregulated in *Acomys* but not in *Mus* (116) (Figure 4A, 4D and 4E). Only 1 gene was downregulated in both species (Figure 4E). When these gene sets were subjected to Panther analysis, a total of 1648 BP ontological categories were found to be enriched in *Mus*, with the associated genes organized into the following themes: processes related to epidermal structure and morphogenesis (5.6%), immune related processes (2.2%), cell cycle and proliferation related processes (5.6%), apoptosis-related processes (1%) and other processes (88.8%). Only 12 BP categories were enriched for *Acomys* (Figure 4F), 8 of which were related to processes related to epidermal structure and morphogenesis (Figure 4G).

In addition, a BP and KEGG pathway analysis conducted with Novomagic software indicated that the response to the DMBA/TPA treatment at D14 did not yield any enriched BP categories or KEGG pathway involving downregulated genes in either *Mus* or *Acomys.* However, the analysis did reveal a number of BP categories and KEGG pathways significantly enriched for upregulated genes in both species (Supplementary Figure 3). Collectively, the data from the D14 analysis suggest that the response in *Mus* primarily involves processes related to the cell cycle and morphological structures while *Acomys* regulates pathways associated with immune response, apoptosis, and tumor suppression.

Given the differential enrichment of immune related processes, particularly for *Acomys*, we conducted IHC against several infiltration markers (Figure 5A). Although CD45, a pan-leukocyte marker, failed to work in *Acomys* (data not shown), both CD68 and Iba1, which are pan-markers for macrophages revealed different levels of infiltration in *Acomys* vs. *Mus* (Figure 5C, D and E). While these two markers are generally accepted as macrophage markers, it is noteworthy that *Mus* exhibits a higher number of CD68+ cells, while *Acomys* shows higher numbers of IBA1+ cells. This discrepancy suggests that these markers identify different subsents of macrophages in both species. . Furthermore, IHC against the apoptotic marker cleaved-caspase 3 seemed generally higher in Acomys than Mus at D14 (Figure 5B and E).

Transcriptional Profile at D28

Mus and *Acomys* samples at D28 (after initial treatment with DMBA and TPA treatment three times a week from D7 to D28) were compared to untreated samples harvested at D0 (N=4). PCA analysis showed expected clustering of samples for *Mus*, with PCA1 and PCA2 explaining 77.4% and 11.64% of variability in *Mus*. *Acomys* samples showed one sample clustering with D0 samples, although overall, PCA1 and PCA2 explained 54.76% and 16.96% of the variance (Supplementary Figure 1). We queried differentially expressed genes (log2 =2) and p-adj ≤ 0.05. A total of 1217 genes were exclusively upregulated in *Mus*, while only 46 genes were exclusively upregulated in *Acomys.* Only 3 genes were upregulated in both species (Figure 6A, 6B and 6D). Panther analysis revealed that genes upregulated in *Mus* were associated with 644 significantly enriched BP ontological categories significantly enriched, with these genes relating to BPs in the following proportions: processes related to epidermal structure and morphogenesis (4.9%), processes related to cell cycle and proliferation (10.7%), and other processes (84.2%; Figure 6C and 6G). No categories were enriched for genes exclusively upregulated in *Acomys* or upregulated genes shared by both species (Figure 6C). Conversely, our analysis of downregulated genes revealed a total of 1763, and 6 genes exclusively downregulated in *Mus* vs *Acomys, respectively* with no genes downregulated in common, and 1528 and 0 processes exclusively downregulated in *Mus* vs *Acomys,* respectively (again with no processes downregulated in common between both species; Figure 6A, 6D, 6E and 6F). When the set of downregulated genes in *Mus* was analyzed using the Panther pipeline, a total of 1528 BP categories were significantly enriched. The genes associated with these BPs corresponded to biological themes in the following proportions: epidermal structure and morphogenesis (2.8%), processed related to cell cycle and proliferation (0.9%), immune related processes (8.5%), apoptosis-related processes (1.2%), and other processes (86.5%; Figure 6F and 6G). In summary, at D28 after initiation of the treatment, *Mus* continues to show significant differential expression of genes, mostly related to cell cycle control and epidermal structure and morphogenesis, but remarkably, the strong differential expression response seen in *Acomys* at D14 (303 and 116 up and downregulated genes respectively) seems to have declined to 57 and 7 up and downregulated genes, respectively, suggesting a return of gene expression almost to baseline levels in *Acomys*. Furthermore, a BP and KEGG pathway analysis conducted with Novomagic software shows that response to the DMBA/TPA insult at D28 does not result in either BP categories or KEGG pathway enrichment involving downregulated genes in either *Mus* or *Acomys* but did reveal several BP categories and KEGG pathways significantly enriched for upregulated genes in both species (Supplementary Figure 4).

Discussion

While *Acomys* is known for its remarkable regenerative properties[31–33,35–37], to date, the relationship between these traits and its incidence and vulnerability to cancer remains unexamined. To investigate the sensitivity or resistance to cancer in *Acomys*, we employed a well-established chemically induced carcinogen regimen. We purposefully chose the C57BL6 strain due to its relatively high level of resistance to papillomas induced by the DMBA/TPA protocol compared to other *Mus* strains[30,44,45]

The number of papillomas we obtained in C57BL6 animals subjected to the DMBA/TPA protocol was in line with what has been published by multiple groups[44–51]. Therefore, for the levels of DMBA and TPA used in the tumor inducing protocol applied, there is a clearly statistically significant difference in the number of papillomas observed 30 weeks into the protocol between Mus and Acomys, suggesting a degree of resistance of *Acomys* to tumorigenesis in the skin. Interestingly, a recent pre-print by White et al, shows similar observations[52].

We attempted to establish whether the initial genotoxic injury with DMBA was comparable in the 2 species by quantifying DSB and proliferation 24 h after treatment by IHC against p2HAx and Ki67: we found that both species had statistically significant increases in the number of p2Hax positive (9.8-fold increase in *Mus* vs. 5.9-fold increase in *Acomys*) and Ki67 positive (3.22-fold increase in *Mus* vs. 1.7-fold increase in *Acomys*) nuclei. In both cases, the fold increases were greater for *Mus* than for *Acomys*, but the number of animals in our experiment did not allow us to determine whether the fold increase differences were statistically significant. However, it is clear that the lack of tumors observed in *Acomys* cannot be attributed to a lack of initial genotoxic injury in *Acomys*. Subsequent to DMBA treatment, TPA treatment was similar throughout the duration of the experiment, after which there was a clear difference in papilloma number. We did not test multiple groups with different concentrations of DMBA and TPA. It is possible that *Acomys* would indeed develop papillomas with higher dosages of DMBA and/or TPA, but regardless, our data clearly show a difference in tumor formation resistance between both species.

Taken together, our data indicate that a) *Acomys* shows resistance to tumor induction using a protocol that clearly induces papillomas in *Mus*; 2) this difference is not due to failure of the treatment to induce DSBs in *Acomys*; 3) both species mount an initial proliferative response; 4) both species undergo immediate infiltration by immune cells; 4) *Acomys* shows higher levels of apoptotic activity compared to Mus at D14 of treatment.

We chose to explore the response of both species to the tumor inducing protocol during the first 28 days of treatment by transcriptomic analysis. The transcriptomic response of *Mus* and *Acomys* to the treatment differ significantly. At D1, *Mus* reacts with differential upregulation of gene expression in a total of 414 genes, while *Acomys* upregulates only 17 genes. The BP categories enriched in *Mus* focus mainly on cell cycle and epidermal structure and morphogenesis. In contrast, the 17 genes upregulated in *Acomys* include upregulation of the detoxification gene *NQ01*, as well as other genes with cancer prevention functions, including the tumor suppressor *SPINK 7*. The somewhat muted response in *Acomys* in terms of number of upregulated and downregulated genes at D1 was unexpected, and it could be argued that this indicates that the initial genotoxic insult was insufficient to trigger the tumorigenic process in *Acomys*. However, two observations counter this argument: first, the level of response to DMBA treatment at D1 in terms of amount of DSB and proliferation was similar in both species and second, the transcriptomic response, in terms of number of DEGs, is similar for both species (479 for *Mus*; 419 for *Acomys*) at D14 after treatment initiation. Interestingly, while the overall number of DEGs at D14 between the species was similar in magnitude, the DEGS were clearly distinct in terms of the functions those genes are involved in.

Our transcriptomic data suggests that by D14, the response of *Mus* to the genotoxic insult focuses on modifications to epidermal structure and an attempt to modulate proliferation and cell cycle related processes, while *Acomys*, in contrast, seems to unleash a multifaceted immune response. This suggests that the molecular strategies against tumorigenesis are different in these two species, and that *Acomys sp* may possess unique mechanisms to resist tumorigenesis, even under conditions that typically induce cancer in non-regenerative mammals. In particular, we found significant upregulation of genes that have been reported to be involved in tumor suppression in a number of contexts in the literature (Table 1 and Table 2). This finding suggests the possibility that tumor suppression may play a role in *Acomys* cancer resistance. However, many tumor suppressor genes are known to be highly context dependent, with some TS acting in specific situations or even being drivers of tumorigenesis. The significance of this finding will need further confirmation.

It is important to recognize that the relationship between enhanced regenerative capabilities and cancer susceptibility is complex, and a deeper mechanistic understanding remains elusive both generally across animal groups and in *Acomys* particularly. While regenerative capacity may confer protection against tumorigenesis, rapid cell proliferation, an essential component of regeneration, can also increase the likelihood of mutations and tumor development. Interestingly, a recent report found significant roadblocks to reprogramming *Acomys* cells, suggesting that tumor suppressor pathways play an important role in *Acomys* regeneration and that *Acomys* may possess unreported cancer resistance[41].

Our results fit the pattern of animal model studies found in the literature associating high regenerative capabilities with cancer resistance. Possibly, the key relation between regeneration and cancer resistance resides in the ability to control and cease proliferation during wound healing in an adaptive manner, and that these mechanisms are supercharged in a species that uses regeneration as a wound healing strategy. However, our transcriptomic analysis suggests that there are several layers to mechanisms deployed to surveil proliferation in *Acomys,* with the immune system playing a key role in controlling tumorigenic processes. This is evident from the observation that *Mus* seems to concentrate its control efforts on modulating the structure and state of the epidermis (keratinocyte proliferation and differentiation) and controlling cell cycle progression, while *Acomys* modulates 29.1% of the genes it upregulates at D14 of treatment in BP categories related to immune response[37,53–61]. In the context of the relationship between regeneration and cancer resistance, our results prompt us to focus on the role of the *Acomys* immune system. While there is consensus in the literature regarding the significance of the *Acomys* immune system in regeneration, and several studies have explored specific aspects[62,63], much remains unknown. *Acomys* shows a blunted immune response involving a decrease in inflammatory cytokines. Gawriluk et al. observed that regeneration was associated with lower levels of pro-inflammatory cytokines (i.e., IL-6, CCL2, and CXCL1) and an increase in local levels of IL-12 and IL-17, correlating with an influx of T-cells into the wound area[63]. On day 14, we did not find differential expressions of IL-6, CCL2, and CXCL1, but identified lower levels of other cytokines considered generally pro-inflammatory, such as IL-12, Il-17, IL-1a and IL-18 in *Acomys*. Also, macrophages have been shown to be required for epimorphic regeneration[62]. Our Panther analysis of BP categories enriched in *Acomys* at D14 found that over 50% of BP categories related to a range of processes related to immunity, including T-cell proliferation, differentiation, migration, and cytokine production, NK chemotaxis and differentiation, macrophage associated processes and a number of cytokine-associated pathways. Our results suggest the immune system of *Acomys* is crucially positioned to determine the outcomes of wound healing, regeneration and cancer resistance. The general characterizations of immune responses in *Acomys* as simply blunt or heightened in comparison to that of non-regenerators seem to be premature and fail to capture the complexity and context-dependent response of the system. A thorough dissection of the *Acomys* immune response in relation to regeneration and cancer resistance is needed.

Methods

Animals

Specimens of *A. dimidiatus* and *Mus musculus* C56BL6 were kept at the animal facility of the Algarve Biomedical Center Research Institute at the University of Algarve, using standard husbandry procedures for each species. Animalswere originally purchased from Charles Rivers. *Acomys* were kept in a room with controlled temperature (26ºC), on a 13/11 h light/dark cycle and fed twice a week with mixed seeds supplemented with fresh fruits and vegetables, with water ad libitum, as described for this species[64]. All experiments were performed in accordance with European guidelines (2010/63/EU) for the care and use of laboratory animals, as well as Portuguese law (DL 113/2013). The project (Study of the Mechanisms of Cancer Resistance in the African Spiny Mouse (Acomys sp) and all experimental procedures were reviewed and approved by the Animal Welfare Body of the University of Algarve and meet the requirements of the Decreto-Lei 113/2013 and the Despacho 2880/2015 of the Direcção Geral de Alimentação e Veterinária of Portugal.

DMBA/TPA treatment

To evaluate the induction of tumors by DMBA/TPA two stage protocol, the dorsal hair on a square of 2 x 2 cm on the back of *Mus musculus* C57BL/6N and *Acomys dimidiatus* animals (N = 6; 1:1 male/female ratio) was trimmed and the skin was treated topically with DMBA (Sigma-Aldrich; 100 µg in 200 µL acetone). *Mus* animals were between 7 and 9 week of age, while *Acomys* animals were approximately 3.5 months of age. The solution was applied evenly throughout the surface of the trimmed area. Starting one week after DMBA application, animals were treated twice weekly with an evenly applied solution of TPA (Sigma-Aldrich; 6.25 μg in 200 μL absolute ethanol) on the trimmed surface for 30 weeks. Tumor formation was assessed weekly. For a transcriptomic analysis of events during the first 28 days of treatment, *Mus* and *Acomys* animals were subjected to the same tumor inducing protocol and samples harvested as described above at day 1 (D1, 24 hours after treatment with DMBA), day 14 (D14, after initial treatment with DMBA and TPA treatment on D7, D9 and D11) and day 28 (D28, after initial treatment with DMBA and TPA treatment 3 times a week from D7 to D28). Harvest involved humanely sacrificing the animals (anesthesia with isofluorane followed by cervical dislocation) and collecting the entire 2 x 2 cm^2 surface area treated with DMBA/TPA.

Immunohistochemistry

Immunohistochemistry was performed in the Histopathology Core Facility at the Institute for Research in Biomedicine in Barcelona (Spain), following standard protocols. Briefly, *Acomys* dorsal skin was fixed in 4% paraformaldehyde overnight, embedded in paraffin, and cut into 3-5 μm sections. For immunostaining, the sections were dewaxed and epitope retrieval was performed with ER1 buffer (AR9961, Leica Biosystems) for Ki67 (Abcam, 15580) for 30 min and with ER2 buffer (AR9640, Leica Biosystems) for Cleaved-Caspase 3 (Cell Signaling, 9661), CD45 (Cell Signaling, 98819), H2AX (Cell Signaling, 9718) and CD68 (Byorbit, orb47985) for 20 min. Washings were performed using the BOND Wash Solution 10x (AR9590, Leica). Quenching of endogenous peroxidase was performed by 10 min of incubation with Peroxidase-Blocking Solution at RT (S2023, Dako, Agilent). Non-specific unions were blocked using 5 % of goat normal serum (16210064, Life technology) mixed with 2.5 % BSA diluted in wash buffer for 60 min at RT. Secondary antibody used was the BrightVision Poly-HRP-Anti Rabbit IgG Biotin-free, ready to use (DPVR-110HRP, Immunologic). Antigen–antibody complexes were reveled with the DAB (Polymer) (Leica, RE7230CE). Sections were counterstained with hematoxylin (RE7107, Leica Biosystems) and mounted with Mounting Medium, Toluene-Free (CS705, Dako, Agilent) using a Dako CoverStainer. For Iba1 (019-19741, Wako), samples were dewaxed and antigen retrieval process using citrate buffer pH6 for 20 min at 97ºC using a PT Link (Dako – Agilent) was performed. Blocking was performed with Peroxidase-Blocking Solution at RT (S2023, Agilent) and 5 % of goat normal serum (16210064, Life technology) mixed with 2.5 % BSA diluted in wash buffer for 10 and 60 min at RT. The secondary antibody used was the BrightVision poly HRP-Anti-Rabbit IgG, incubated for 45min (DPVR-110HRP, ImmunoLogic). Antigen–antibody complexes were reveled with 3-3'-diaminobenzidine (K346811, Agilent). Sections were counterstained with hematoxylin (CS700, Dako, Agilent) and mounted with Mounting Medium, Toluene-Free (CS705, Agilent) using a Dako CoverStainer. Specificity of staining was confirmed staining with the rabbit IgG, polyclonal (NBP2-24891, Novus bio-tec). Digital scanned brightfield images were acquired with a NanoZoomer-2.0 HT C9600 scanner (Hamamatsu, Photonics, France) equipped with a 20X objective and using NDP.scan2.5 software U10074-03 (Hamamatsu, Photonics, France). All images were visualized with the NDP.view 2 U123888-01 software (Hamamatsu, Photonics, France) with a gamma correction set at 1.8 in the image control panel of the NDP.view 2

U123888-01 software (Hamamatsu, Photonics, France).. Immunohistochemical (IHC) number of positive cell determination was performed using QuPath software (version 0.5.0) on a total of 3 animals per experimental group. For each animal, one entire tissue section was analyzed using the automated Positive Cell Detection tool after defining the image type as Brightfield H-DAB25. This method allows for the detection and quantification of positively stained cells across the entire tissue sample.

RNA sequencing

Total RNA was extracted from mice and *Acomys* skin tissues using Tryzol (Nzytech). Briefly, skin was first homogenized using Navy lysis kit (Next Advance) and extracted using Tryzol (Nzytech). RNA integrity was assessed using the Bioanalyzer 2100 system (Agilent Technologies). Messenger RNA was enriched from total RNA using poly-T oligo-attached magnetic beads. After fragmentation, the first strand cDNA was synthesized using random hexamer primers, followed by the second strand cDNA synthesis using dUTP. Libraries were subjected to end repair, A-tailing, adapter ligation, size selection, amplification, and purification. The libraries were checked with Qubit and real-time PCR for quantification and Bioanalyzer 2100 system (Agilent Technologies) for size distribution detection. After library quality control, different libraries were pooled based on the effective concentration and targeted data amount, then sequenced by Novogene Europe on the NovaSeq™ X Plus platform (Illumina).

Raw data (raw reads) of fastq format were processed through fastp software. In this step, clean data (clean reads) were obtained by removing reads containing adapters, reads containing poly-N and low-quality reads from raw sequence. At the same time, Q20, Q30 and GC content of the clean data were calculated. All the downstream analyses were based on high quality, clean reads. The trimmed reads were mapped to either the *Mus musculus* (GCA_947599735.1) or *Acomys dimidiatus* (GCA_907164435.1) reference genomes using HISAT2 v2.05 software. We obtained over 20,000 reads for each species. Because the *Acomys* genome is incompletely annotated, we limited our downstream analysis to a subset of 16,542 genes which showed an unequivocal 1 to 1 ortholog relationship between both species, using *Mus* related databases to complete the analysis. The mapped reads of each sample were assembled by StringTie (v1.3.3b) (Pertea, 2015) in a reference-based approach. FeatureCounts v1.5.0-p3 was used to count the read numbers mapped to each gene and then FPKM of each gene was calculated based on the length of the gene and reads count mapped to this gene. Prior to differential gene expression analysis, for each sequencing library, read counts were adjusted using the edgeR R package (3.22.5) by scaling normalization factors to eliminate differences in sequencing depth between samples. Differential expression analysis for two conditions/groups was performed using the DESeq2 R package (1.20.0). The resulting p-value was adjusted using the Benjamini and Hochberg's methods to control the error discovery rate. The corrected p-value ≤ 0.05 & log2 (foldchange)| was set at 1 for D1, and 2 for D14 and D28, as the threshold of significant differential expression. GO terms, Kegg pathways were analyzed using Panther (https://www.pantherdb.org) and NovoMagic (Novogene) bioinformatic software.

This study has been reported in accordance to ARRIVAL guidelines, except for preregistration of the study.

Declarations

Acknowledgements

This work was supported by a grant from Spanish Ministerio de Ciencia e Innovación/Agencia Estatal de Investigación and the European Regional Development Fund (PID2022-136654OB-I00 financed by MCIN/AEI /10.13039/501100011033 / FEDER, UE). This study received Portuguese national funds from FCT - Foundation for Science and Technology through projects UIDB/04326/2020 (DOI:10.54499/UIDB/04326/2020), UIDP/04326/2020 (DOI:10.54499/UIDP/04326/2020) and LA/P/0101/2020 (DOI:10.54499/LA/P/0101/2020).

We acknowledge the expertise and dedication of the Animal House of ABC Ri-UAlg.

Author contributions

MV, IG and GP designed and executed experiments; MV, BF, WL and IA analyzed the results; WL and GT supervised the project, analyzed data and wrote the manuscript.

Data availability statement

Correspondence and requests for materials should be addressed to Gustavo Tiscornia. Supplementary Information is available for this paper. Reprints and permissions information is available at www.nature.com/reprints.

Conflicts of Interest

The authors declare no conflict of interest. Wolfgang Link is the scientific co-founder of Refoxy Pharmaceuticals GmbH, Cologne and is required by his institution to state so in his publications. The funders had no role in the design and writing of the manuscript.

References

1. Albuquerque, T. A. F., Drummond do Val, L., Doherty, A. & de Magalhães, J. P. From humans to hydra: patterns of cancer across the tree of life. *Biological Reviews* **93**, 1715–1734 (2018).
2. Caulin, A. F. & Maley, C. C. Peto's Paradox: evolution's prescription for cancer prevention. *Trends Ecol Evol* **26**, 175–182 (2011).
3. Kristine P. Krafts. Tissue Repair: The hidden drama. *Organogenesis* **6**, 225233 (2010).
4. Rybinski, B., Franco-Barraza, J. & Cukierman, E. The wound healing, chronic fibrosis, and cancer progression triad. *Physiol Genomics* **46**, 223–244 (2014).
5. Schäfer, M. & Werner, S. Cancer as an overhealing wound: an old hypothesis revisited. *Nat Rev Mol Cell Biol* **9**, 628–638 (2008).

6. Maggiore, G. & Zhu, H. Relationships Between Regeneration, Wound Healing, and Cancer. *Annu Rev Cancer Biol* **8**, 177–197 (2024).
7. Arnold, K. M., Opdenaker, L. M., Flynn, D. & Sims-Mourtada, J. Wound Healing and Cancer Stem Cells: Inflammation as a Driver of Treatment Resistance in Breast Cancer. *Cancer Growth Metastasis* **8**, CGM.S11286 (2015).
8. Foster, D. S., Jones, R. E., Ransom, R. C., Longaker, M. T. & Norton, J. A. The evolving relationship of wound healing and tumor stroma. *JCI Insight* **3**, (2018).
9. Sundaram, G. M., Quah, S. & Sampath, P. Cancer: the dark side of wound healing. *FEBS J* **285**, 4516–4534 (2018).
10. Flier, J. S., Underhill, L. H. & Dvorak, H. F. Tumors: Wounds That Do Not Heal. *New England Journal of Medicine* **315**, 1650–1659 (1986).
11. Dvorak, H. F. Tumors: Wounds That Do Not Heal—Redux. *Cancer Immunol Res* **3**, 1–11 (2015).
12. Wong, A. Y. & Whited, J. L. Parallels between wound healing, epimorphic regeneration and solid tumors. *Development* **147**, (2020).
13. Reddien, P. W. The Cellular and Molecular Basis for Planarian Regeneration. *Cell* **175**, 327–345 (2018).
14. Rink, J. C. Stem cell systems and regeneration in planaria. *Dev Genes Evol* **223**, 67–84 (2013).
15. Stephan, F. [Spontaneous tumors in the planarian Dugesia tigrina]. *C R Seances Soc Biol Fil* **156**, 920–2 (1962).
16. Voura, E. B. *et al.* Planarians as models of cadmium-induced neoplasia provide measurable benchmarks for mechanistic studies. *Ecotoxicol Environ Saf* **142**, 544–554 (2017).
17. Hall, F., Morita, M. & Best, J. B. Neoplastic transformation in the planarian: I. Cocarcinogenesis and histopathology. *Journal of Experimental Zoology* **240**, 211–227 (1986).
18. Plusquin, M. *et al.* Physiological and molecular characterisation of cadmium stress in <I>Schmidtea mediterranea</I>. *Int J Dev Biol* **56**, 183–191 (2012).
19. Pearson, B. J. & Alvarado, A. S. A planarian p53 homolog regulates proliferation and self-renewal in adult stem cell lineages. *Development* **137**, 213–221 (2010).
20. Oviedo, N. J., Pearson, B. J., Levin, M. & Sánchez Alvarado, A. Planarian PTEN homologs regulate stem cells and regeneration through TOR signaling. *Dis Model Mech* **1**, 131–143 (2008).
21. Harshbarger, J. C. *et al.* Cutaneous mastocytomas in the neotenic caudate amphibians Ambystoma mexicanum (axolotl) and Ambystoma tigrinum (tiger salamander). *J Cancer Res Clin Oncol* 187–192 (1999).
22. Ingram, A. J. The reactions to carcinogens in the axolotl (Ambystoma mexicanum) in relation to the 'regeneration field control' hypothesis. *J Embryol Exp Morphol* **26**, 425–41 (1971).
23. Gemberling, M., Bailey, T. J., Hyde, D. R. & Poss, K. D. The zebrafish as a model for complex tissue regeneration. *Trends in Genetics* **29**, 611–620 (2013).
24. Liu, S. & Leach, S. D. Zebrafish models for cancer. *Annu Rev Pathol* **6**, 71–93 (2011).

25. White, R., Rose, K. & Zon, L. Zebrafish cancer: the state of the art and the path forward. *Nat Rev Cancer* **13**, 624–636 (2013).
26. Demirci, Y. *et al.* Brain Regeneration Resembles Brain Cancer at Its Early Wound Healing Stage and Diverges From Cancer Later at Its Proliferation and Differentiation Stages. *Front Cell Dev Biol* **10**, (2022).
27. Hale, A. J., Kiai, A., Sikkens, J. & den Hertog, J. Impaired caudal fin-fold regeneration in zebrafish deficient for the tumor suppressor Pten. *Regeneration* **4**, 217–226 (2017).
28. Li, H. & Brakebusch, C. Analyzing skin tumor development in mice by the DMBA/TPA model. in *Methods Cell Biol* . vol. 163 113–121 (2021).
29. Vähätupa, M., Pemmari, T., Junttila, I., Pesu, M. & Järvinen, T. A. H. Chemical-Induced Skin Carcinogenesis Model Using Dimethylbenz[a]Anthracene and 12-O-Tetradecanoyl Phorbol-13-Acetate (DMBA-TPA). *Journal of Visualized Experiments* (2019) doi:10.3791/60445.
30. Abel, E. L., Angel, J. M., Kiguchi, K. & DiGiovanni, J. Multi-stage chemical carcinogenesis in mouse skin: Fundamentals and applications. *Nat Protoc* **4**, 1350–1362 (2009).
31. Matias Santos, D. *et al.* Ear wound regeneration in the African spiny mouse Acomys cahirinus . *Regeneration* **3**, 52–61 (2016).
32. Maden, M. & Brant, J. Insights into the regeneration of skin from Acomys , the spiny mouse. *Exp Dermatol* **28**, (2018).
33. Maden, M. *et al.* Perfect chronic skeletal muscle regeneration in adult spiny mice, Acomys cahirinus. *Sci Rep* **8**, 8920 (2018).
34. Gawriluk, T. R. *et al.* Comparative analysis of ear-hole closure identifies epimorphic regeneration as a discrete trait in mammals. *Nat Commun* **7**, 11164 (2016).
35. Nogueira-Rodrigues, J. *et al.* Rewired glycosylation activity promotes scarless regeneration and functional recovery in spiny mice after complete spinal cord transection. *Dev Cell* **57**, 440-450.e7 (2022).
36. Okamura, D. M., Nguyen, E. D., Beier, D. R. & Majesky, M. W. Wound healing and regeneration in spiny mice (Acomys cahirinus). in *Curr Top Dev Biol* . vol. 148 139–164 (2022).
37. Seifert, A. W. *et al.* Skin shedding and tissue regeneration in African spiny mice (Acomys). *Nature* **489**, 561–565 (2012).
38. Sandoval, A. G. W. & Maden, M. Regeneration in the spiny mouse, Acomys, a new mammalian model. *Curr Opin Genet Dev* **64**, 31–36 (2020).
39. Koopmans, T. *et al.* Ischemic tolerance and cardiac repair in the spiny mouse (Acomys). *NPJ Regen Med* **6**, 78 (2021).
40. Okamura, D. M. *et al.* Spiny mice activate unique transcriptional programs after severe kidney injury regenerating organ function without fibrosis. *iScience* **24**, 103269 (2021).
41. Sandoval, A. G. W., Maden, M., Bates, L. E. & Silva, J. C. R. Tumor suppressors inhibit reprogramming of African spiny mouse (Acomys) fibroblasts to induced pluripotent stem cells. *Wellcome Open Res*

7, 215 (2022).

42. Oh, E.-T. & Park, H. J. Implications of NQO1 in cancer therapy. *BMB Rep* **48**, 609–617 (2015).
43. Androutsopoulos, V. P., Tsatsakis, A. M. & Spandidos, D. A. Cytochrome P450 CYP1A1: wider roles in cancer progression and prevention. *BMC Cancer* **9**, 187 (2009).
44. Slaga, T. J. SENCAR mouse skin tumorigenesis model versus other strains and stocks of mice. *Environ Health Perspect* **68**, 27–32 (1986).
45. Hennings, H. *et al.* FVB/N mice: an inbred strain sensitive to the chemical induction of squamous cell carcinomas in the skin. *Carcinogenesis* **14**, 2353–2358 (1993).
46. Oka, K. *et al.* Resistance to chemical carcinogenesis induction via a dampened inflammatory response in naked mole-rats. *Commun Biol* **5**, 287 (2022).
47. Yan, B. *et al.* Increased skin carcinogenesis in caspase-activated DNase knockout mice. *Carcinogenesis* **30**, 1776–1780 (2009).
48. Swann, J. B. *et al.* Demonstration of inflammation-induced cancer and cancer immunoediting during primary tumorigenesis. *Proceedings of the National Academy of Sciences* **105**, 652–656 (2008).
49. Vähätupa, M., Pemmari, T., Junttila, I., Pesu, M. & Järvinen, T. A. H. Chemical-Induced Skin Carcinogenesis Model Using Dimethylbenz[a]Anthracene and 12-O-Tetradecanoyl Phorbol-13-Acetate (DMBA-TPA). *Journal of Visualized Experiments* (2019) doi:10.3791/60445.
50. Zeng, F. *et al.* Interleukin-37 promotes DMBA/TPA skin cancer through SIGIRR-mediated inhibition of glycolysis in CD103 $^{+}$ DC cells. *MedComm (Beijing)* **4**, (2023).
51. Sundberg, J. P., Sundberg, B. A. & Beamer, W. G. Comparison of chemical carcinogen skin tumor induction efficacy in inbred, mutant, and hybrid strains of mice: Morphologic variations of induced tumors and absence of a papillomavirus cocarcinogen. *Mol Carcinog* **20**, 19–32 (1997).
52. White, A., Donahue, L., Hsu, C.-H. & Yang, Y. Skin regeneration underlies cancer resistance in Acomys. Preprint at https://doi.org/10.21203/rs.3.rs-5160289/v1 (2024).
53. McGann, C. J., Odelberg, S. J. & Keating, M. T. Mammalian myotube dedifferentiation induced by newt regeneration extract. *Proceedings of the National Academy of Sciences* **98**, 13699–13704 (2001).
54. Levi, B. P. & Morrison, S. J. Stem Cells Use Distinct Self-renewal Programs at Different Ages. *Cold Spring Harb Symp Quant Biol* **73**, 539–553 (2008).
55. Pajcini, K. V., Corbel, S. Y., Sage, J., Pomerantz, J. H. & Blau, H. M. Transient Inactivation of Rb and ARF Yields Regenerative Cells from Postmitotic Mammalian Muscle. *Cell Stem Cell* **7**, 198–213 (2010).
56. Imokawa, Y. & Brockes, J. P. Selective Activation of Thrombin Is a Critical Determinant for Vertebrate Lens Regeneration. *Current Biology* **13**, 877–881 (2003).
57. Tanaka, E. M., Drechsel, D. N. & Brockes, J. P. Thrombin regulates S-phase re-entry by cultured newt myotubes. *Current Biology* **9**, 792–799 (1999).

58. Godwin, J. W. & Brockes, J. P. Regeneration, tissue injury and the immune response. in *Journal of Anatomy* vol. 209 423–432 (2006).
59. King, M. W., Neff, A. W. & Mescher, A. L. The Developing *Xenopus* Limb as a Model for Studies on the Balance between Inflammation and Regeneration. *Anat Rec* **295**, 1552–1561 (2012).
60. Mescher, A. L. & Neff, A. W. Regenerative Capacity and the Developing Immune System. *Adv Biochem Eng Biotechnol* 39–66 (2005) doi:10.1007/b99966.
61. Calve, S., Odelberg, S. J. & Simon, H.-G. A transitional extracellular matrix instructs cell behavior during muscle regeneration. *Dev Biol* **344**, 259–271 (2010).
62. Simkin, J., Gawriluk, T. R., Gensel, J. C. & Seifert, A. W. Macrophages are necessary for epimorphic regeneration in African spiny mice. *Elife* **6**, (2017).
63. Gawriluk, T. R. *et al.* Complex Tissue Regeneration in Mammals Is Associated With Reduced Inflammatory Cytokines and an Influx of T Cells. *Front Immunol* **11**, (2020).
64. Pinheiro, G., Prata, D. F., Araújo, I. M. & Tiscornia, G. The African spiny mouse (*Acomys* spp.) as an emerging model for development and regeneration. *Lab Anim* **52**, 565–576 (2018).
65. Pertea, M. *et al.* StringTie enables improved reconstruction of a transcriptome from RNA-seq reads. *Nat Biotechnol* **33**, 290–295 (2015).

Tables

Tables 1 to 2 are available in the Supplementary Files section

Figures

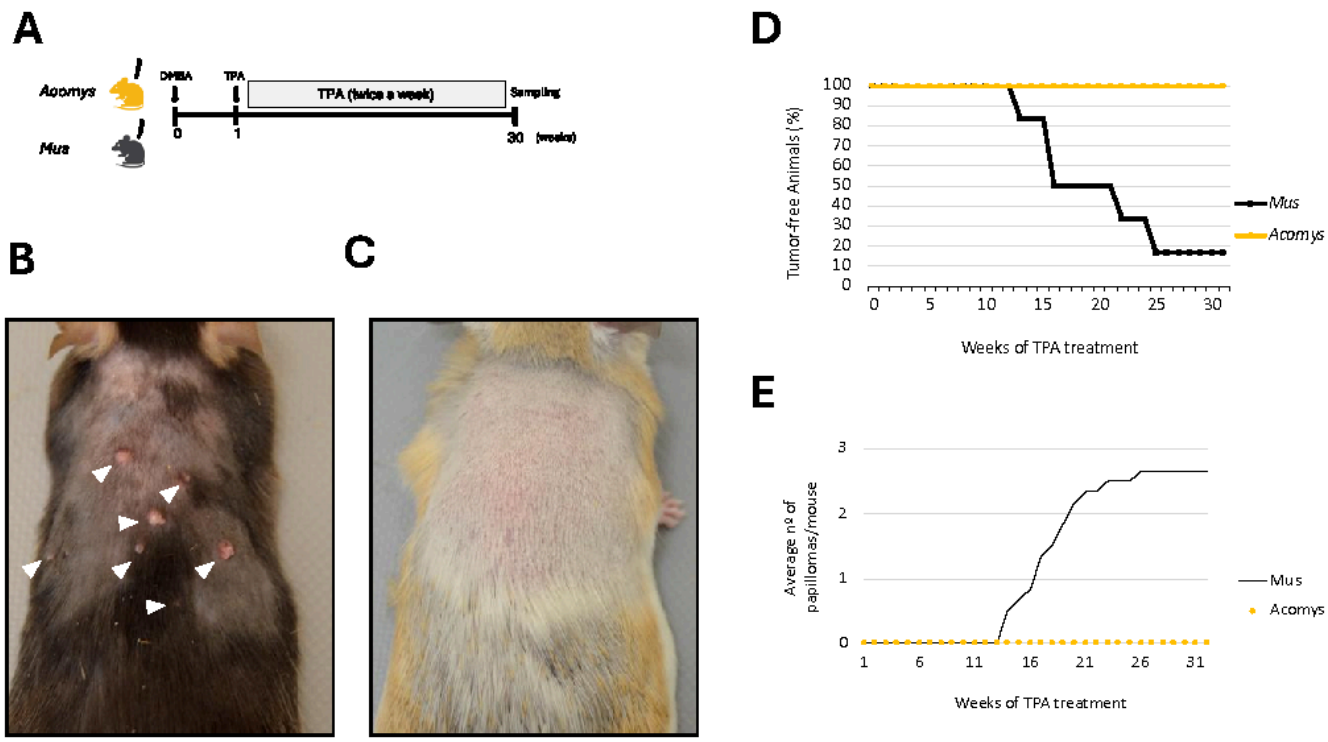

Figure 1 – *Acomys dimidiatus* is resistant to tumor development induced by DMBA/TPA two stage protocol. **(A)** Schematic diagram for carcinogenesis induction by DMBA/TPA treatment on the back skin of both *A. dimidiatus* and *M. musculus*. **(B-C)** Back dorsal skin of *M. musculus* and *A. dimidiatus* 30 weeks after DMBA/TPA treatment with papillomas visible in *M. musculus* **(B)** and absent from *A. dimidiatus* **(C)**. **(D)** Graphical representation of papilloma free animals and (**E**) average number of papillomas per animal over time during DMBA/TPA treatment.

Figure 1

See image above for figure legend.

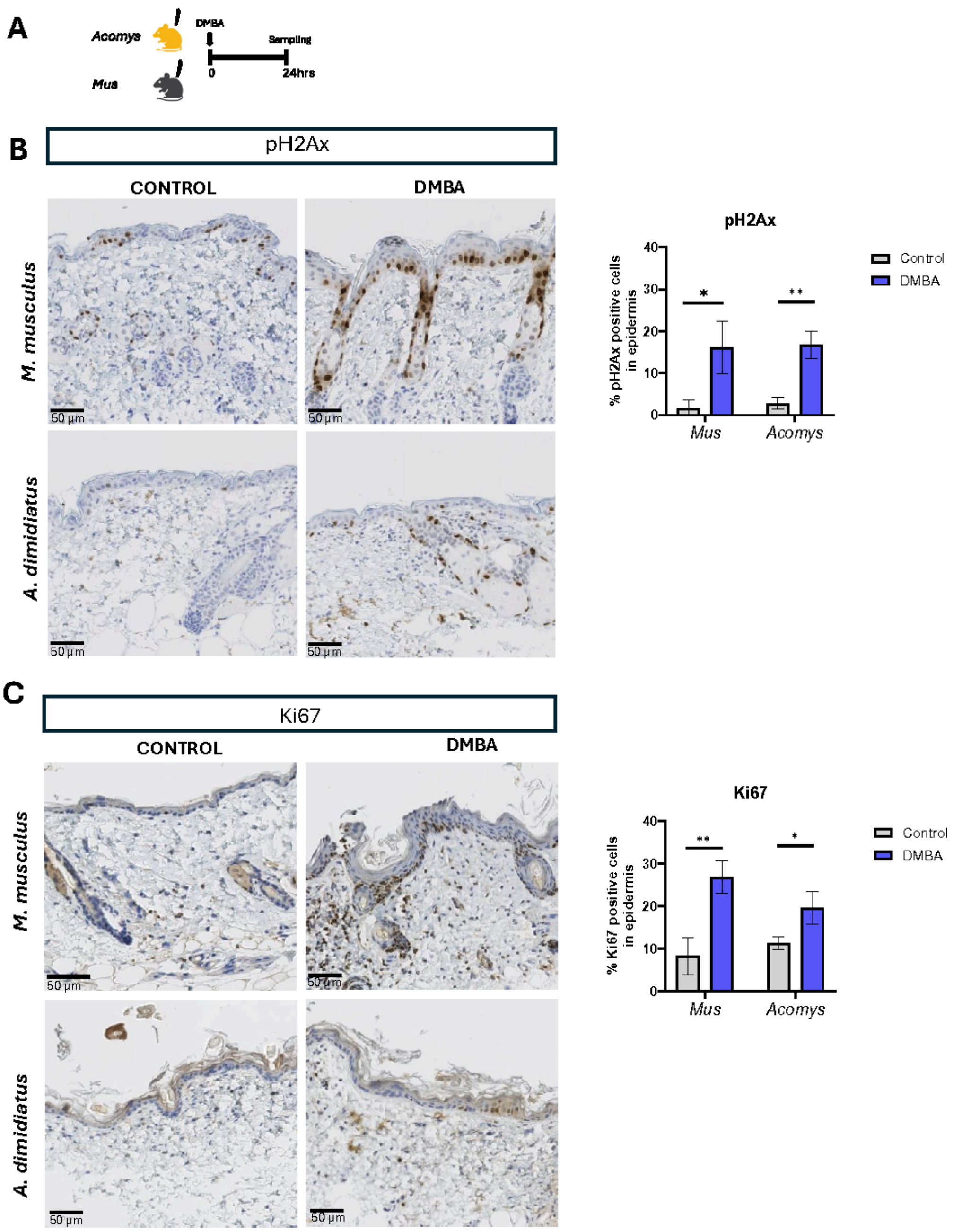

Figure 2 – DMBA triggers double-strand-break formation in both *A. dimidiatus* and *M. musculus*. (A) Schematic illustration depicting the experimental approach to investigate short-term responses to DMBA treatment of both species. **(B-C)** immunohistochemistry staining and quantification of **(B)** pH2Ax-positive cells and **(C)** Ki67- positive cells in *M. musculus* and *A. dimidiatus* one day after DMBA application. Scale bar: 50 µm.

Figure 2

See image above for figure legend.

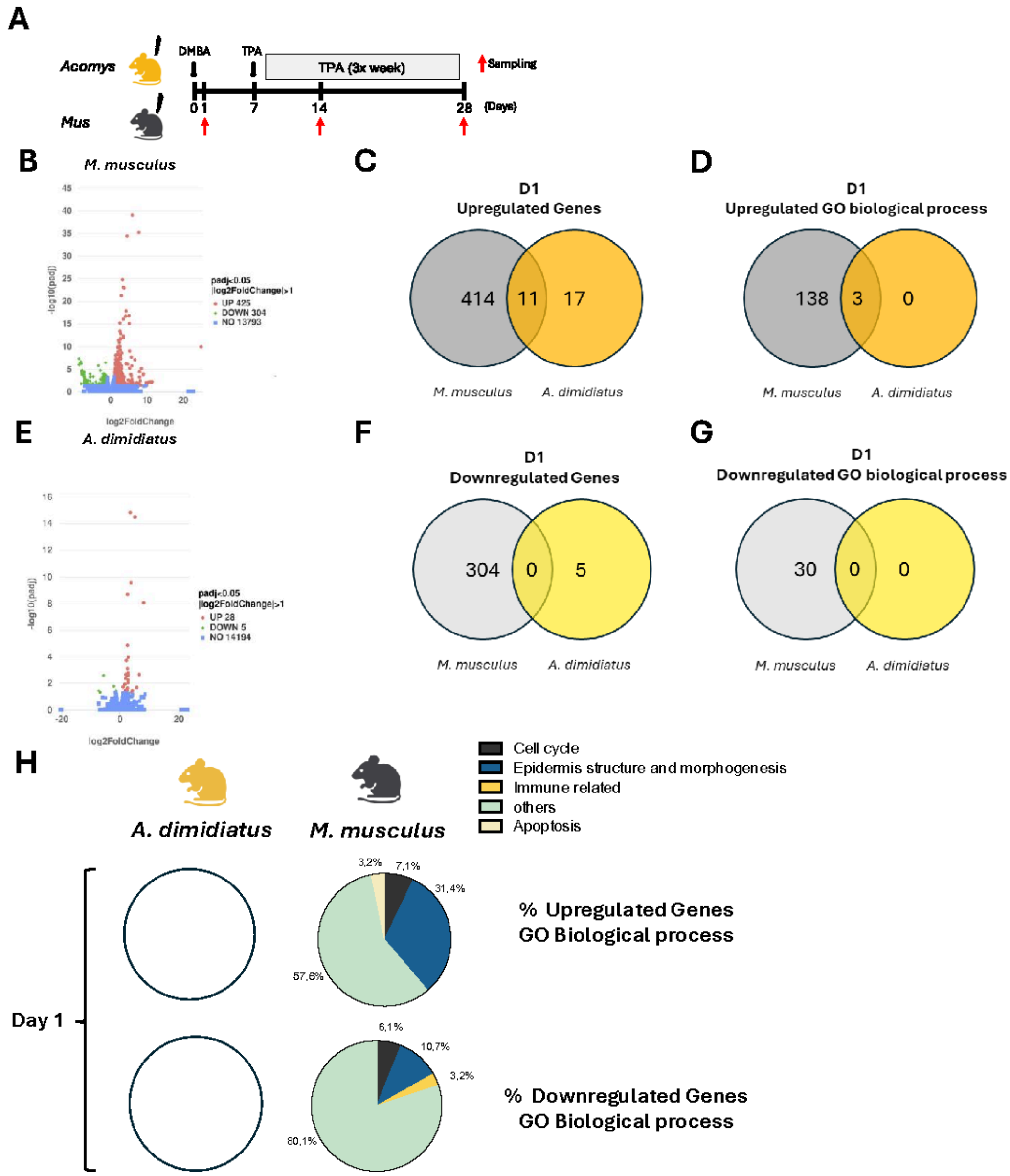

Figure 3 – Administration of DMBA does not induce major differences in the transcriptomic profiles. **(A)** Schematic illustration of transcriptomic experimental design. Red arrow indicates the times of sample collection. Volcano plots of genes differentially expressed in *M. musculus*. **(B, E)** Volcano plots of genes differentially expressed in *M. musculus* **(B)** and *A. dimidiatus* **(E)** one day after treatment with DMBA (D1). The log2 fold change (FC) indicates the mean expression level for each gene. Genes were scored as differentially expressed when log2 FC>1, p<0,05. Each dot represents one gene. Red dots represent upregulate genes, green dots represent downregulated genes, and blue dots represent genes that are not differentially expressed. Venn Diagrams of upregulated genes **(C)** and GO biological processes **(D)**, downregulated genes **(F)** and biological processes **(G)** in both species. **(H)** Percentage of upregulated and downregulated genes included in categories of GO biological processes. White circles are shown when there were no differentially expressed genes identified.

Figure 3

See image above for figure legend.

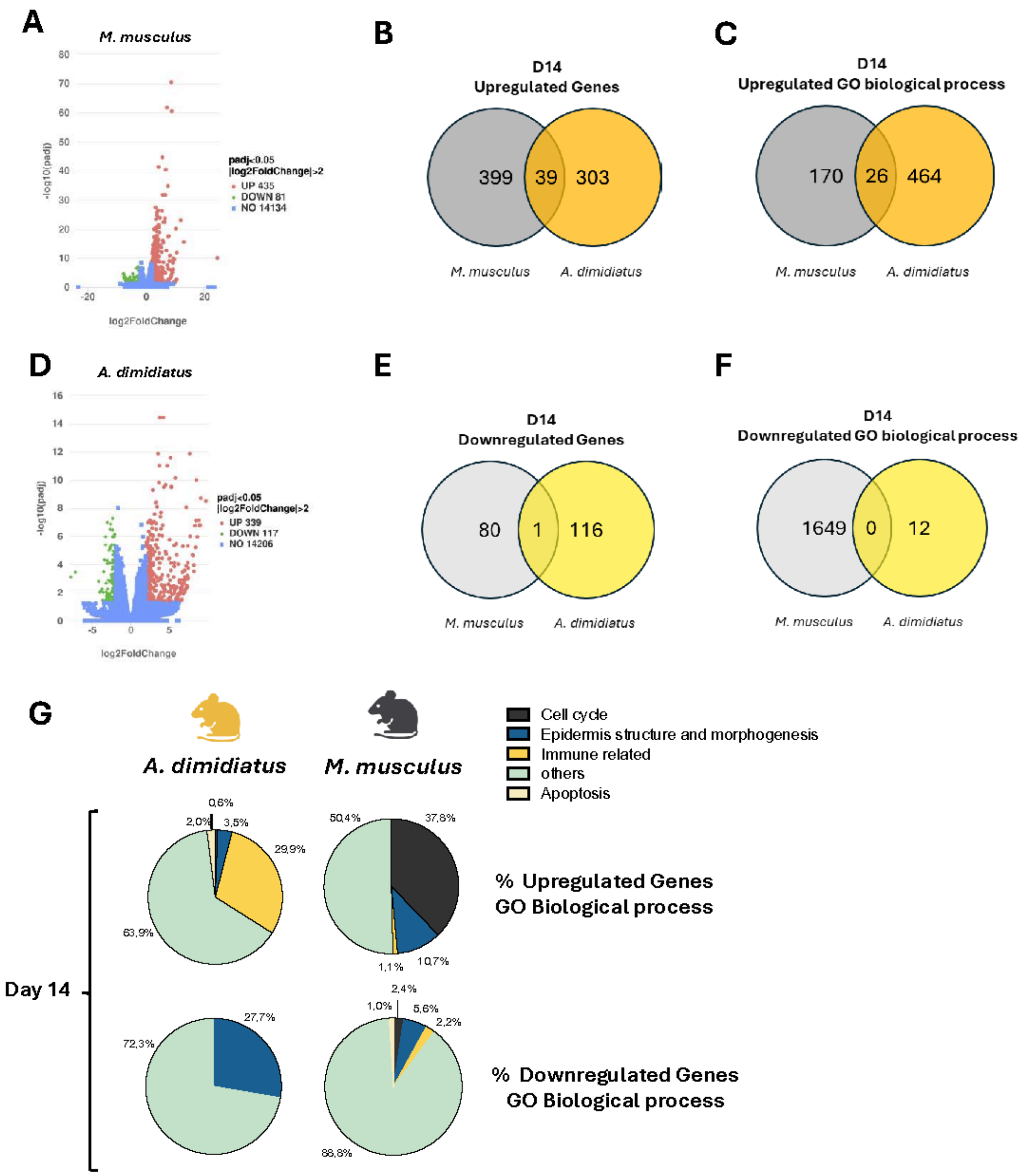

Figure 4: DMBA/TPA treatment induces activation of immune related genes in *A. dimidiatus* versus cell cycle regulation in *M. musculus*. (A) and *A. dimidiatus* **(D)** 14 days after treatment with DMBA (D14). The log2 fold change (FC) indicates the mean expression level for each gene. Genes were scored as differentially expressed when log2 FC>2, p<0,05. Each dot represents one gene. Red dots represent upregulate genes, green dots represent downregulated genes, and blue dots represent genes that are not differentially expressed. Venn Diagrams of upregulated genes **(B)** and GO biological processes **(C)**, downregulated genes **(E)** and biological processes **(F)** in both species. **(G)** Percentage of upregulated and downregulated genes included in categories of GO biological processes

Figure 4

See image above for figure legend.

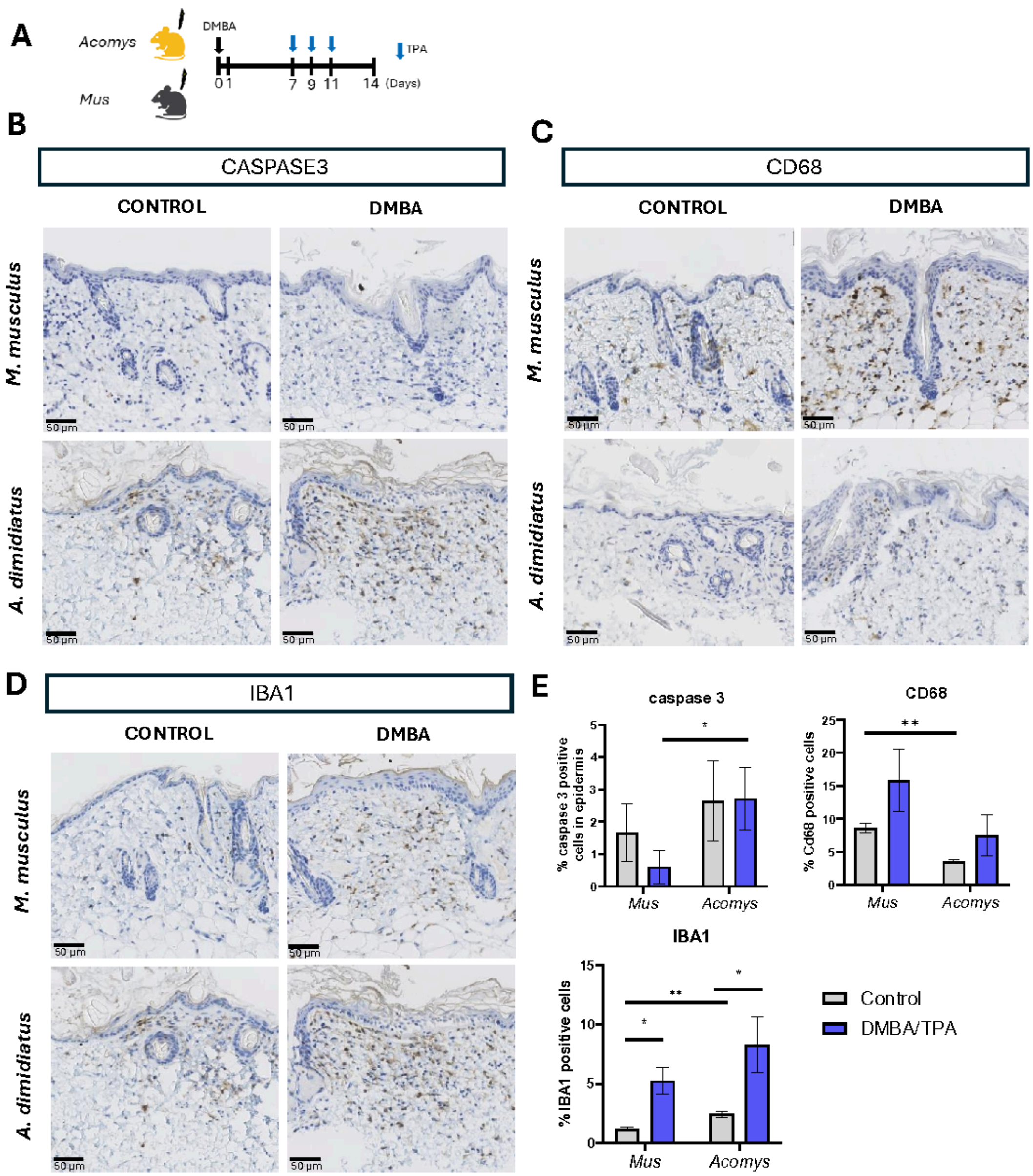

Figure 5 – DMBA induces expression of CD68 in *M. musculus* and IBA1 in *A. dimidiatus*, as well as cleaved-Caspase 3 in *A. dimidiatus* at D14. (A) Schematic illustration of the experimental approach to investigate short-term responses to DMBA treatment of both species. **(B-C)** immunohistochemistry staining of **(B)** cleaved-CASPASE3-positive cells, **(C)** CD68-positive cells and **(D)** IBA1-positive cells in *M. musculus* and *A. dimidiatus* at D14 after DMBA/TPA application. **(E)** Quantification of cleaved-caspase 3, CD68 and IBA1 positive cells in both *M. musculus* and *A. dimidiatus*. Scale bar: 50 µm.

Figure 5

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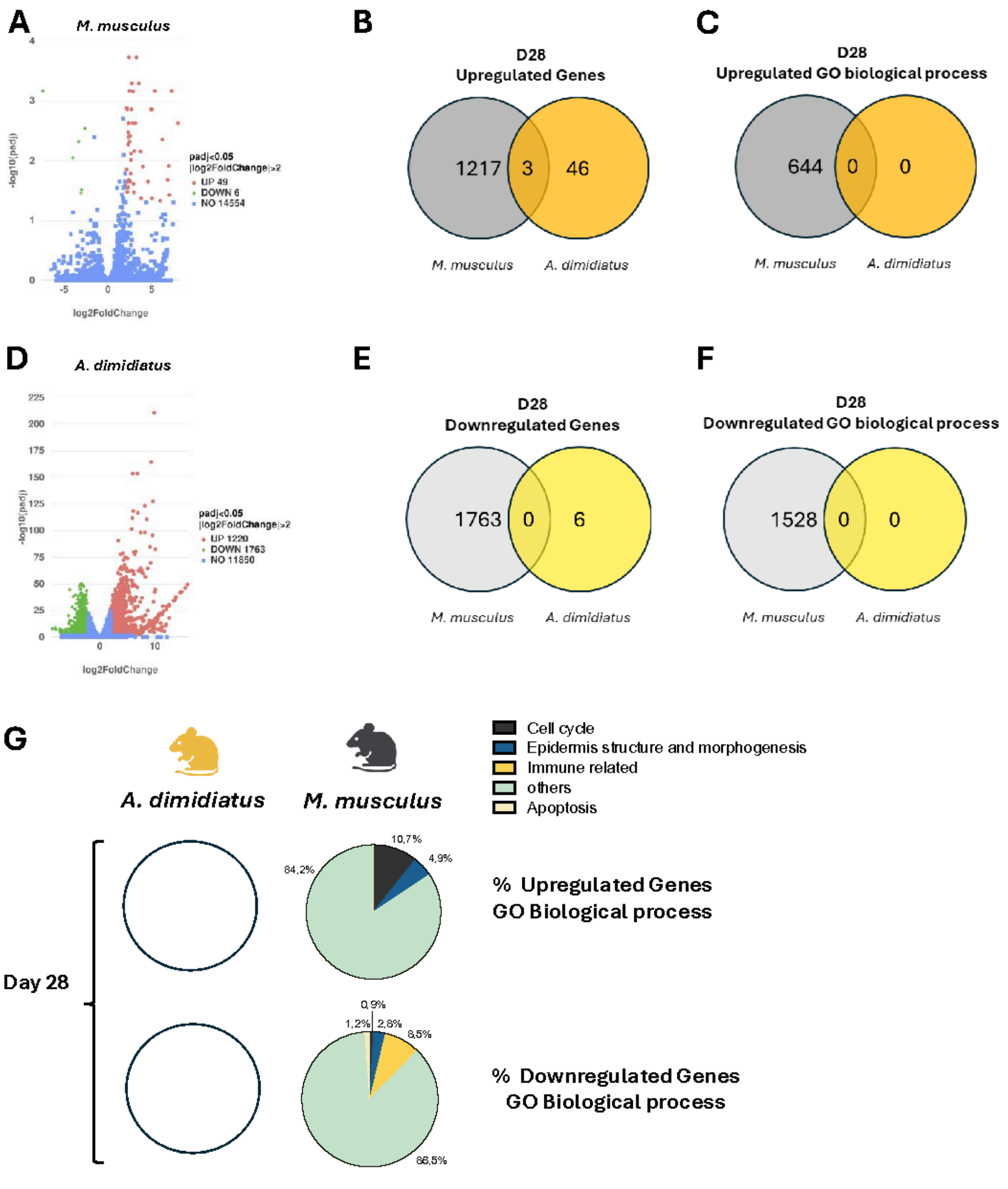

Figure 6: At day 28, *Mus* continues to show significant differential gene expression of genes, but *Acomys* return to baseline. (A, D) Volcano plots of genes differentially expressed in *M. musculus* **(A)** and *A. dimidiatus* **(D)** 28 days after treatment with DMBA (D1). The log2 fold change (FC) indicates the mean expression level for each gene. Genes were scored as differentially expressed when log2 FC>2, p<0,05. Each dot represents one gene. Red dots represent upregulate genes, green dots represent downregulated genes, and blue dots represent genes that are not differentially expressed. Venn Diagrams of upregulated genes **(B)** and GO biological processes **(C)**, downregulated genes **(E)** and biological processes **(F)** in both species. **(G)** Percentage of upregulated and downregulated genes included in categories of GO biological processes. White circles are shown when there were no differentially expressed genes identified.

Figure 6

See image above for figure legend.

Supplementary Files

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